Applied Human Computer Interaction (AHCI)

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Chapter 2 – The Computer

How many computers do you have in your home?

How many computers do you have in your pocket?

Interactivity

Long ago in a galaxy far away … *batch* processing

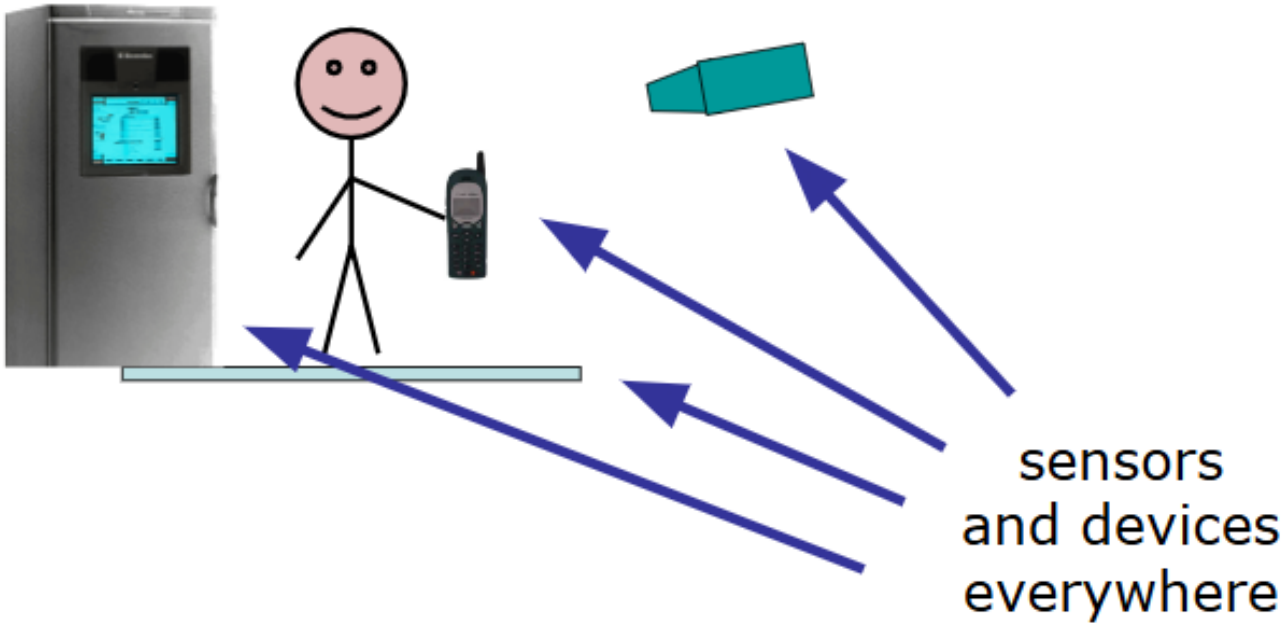
- punched card stacks or large data files prepared
- long wait ….
- line printer output
- … and if it is not right …

Now most computing is interactive

- rapid feedback
- the user in control (most of the time)
- doing rather than thinking …

Is faster always better?

Richer interaction



Text entry devices

Keyboards: primary text entry device

- Most keyboards use a **standardized QWERTY layout**
- QWERTY is **not optimal for typing speed**
- Designed for **mechanical typewriters**
- Prevented key arms from jamming
- Despite better alternatives (DVORAK, alphabetic, chord keyboards):
- QWERTY persists due to **technological & social inertia**
- Users resist relearning
- High cost of replacing hardware

Chord Keyboards

- Use very few keys (4–5 instead of 26+)
- Characters entered by pressing combinations (chords) of keys
- Key combinations often reflect letter shapes (e.g., Microwriter)

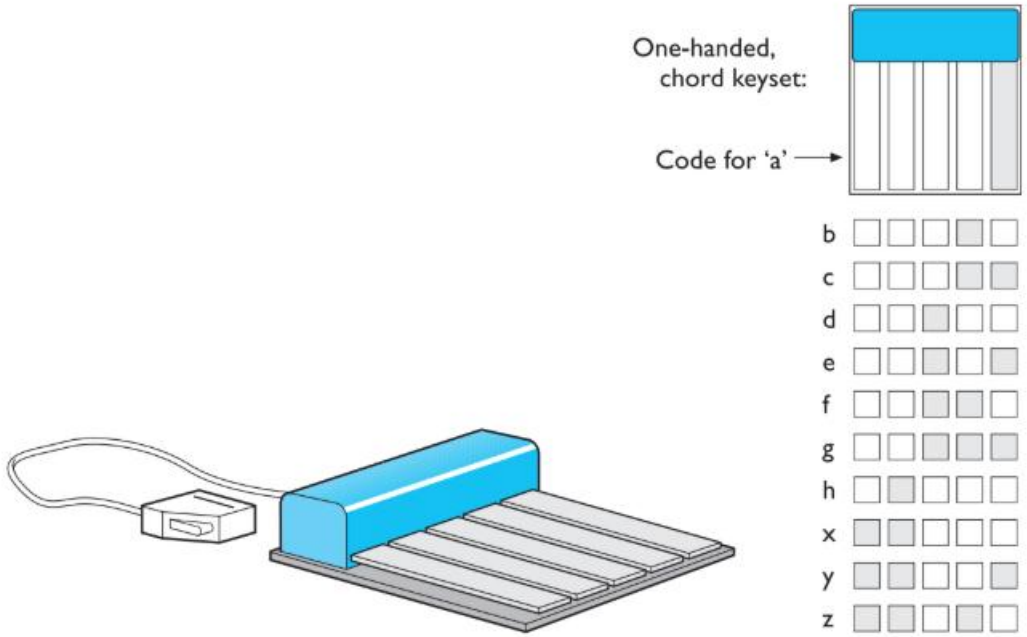


Figure 2.4 A very early chord keyboard (left) and its lettercodes (right)

Design Focus

Numeric keypads

Alphanumeric keyboards (as the name suggests) include numbers as well as letters. In the QWERTY layout these are in a line across the top of the keyboard, but in most larger keyboards there is also a separate number pad to allow faster entry of digits. Number keypads occur in other contexts too, including calculators, telephones and ATM cash dispensers. Many people are unaware that there are two different layouts for numeric keypads: the calculator style that has '123' on the bottom and the telephone style that has '123' at the top.

It is a demonstration of the amazing adaptability of humans that we move between these two styles with such ease. However, if you need to include a numeric keypad in a device you must consider which is most appropriate for your potential users. For example, computer keyboards use calculator-style layout, as they are primarily used for entering numbers for calculations.

One of the authors was caught out by this once when he forgot the PIN number of his cash card. He half remembered the digits, but also his fingers knew where to type, so he 'practiced' on his calculator. Unfortunately ATMs use telephone-style layout!

calculator

ATM

phone

Phone pad and T9 entry

Use numeric keys with multiple presses

- 2 – a b c 6 - m n o 3 - d e f 7 - p q r s
- 4 - g h I 8 - t u v 5 - j k l 9 - w x y z
- hello = 4433555[pause]555666
- surprisingly fast!

T9 predictive entry

- type as if single key for each letter
- use dictionary to ‘guess’ the right word
- hello = 43556 …
- but 26 -> menu ‘am’ or ‘an’
- **Contemporary Example:**
  Early version of: WhatsApp predictive text, Google keyboard suggestions, Auto-complete

Handwriting recognition

- Text can be input into the computer, using a pen and a digesting tablet
- **Technical problems:**
- capturing all useful information - stroke path, pressure, etc. in a natural manner
- segmenting joined up writing into individual letters
- interpreting individual letters
- coping with different styles of handwriting

Speech Recognition

- Speech is a natural & familiar form of interaction
- Systems can achieve high accuracy (~97%)
- Accuracy usually applies only to limited command vocabularies

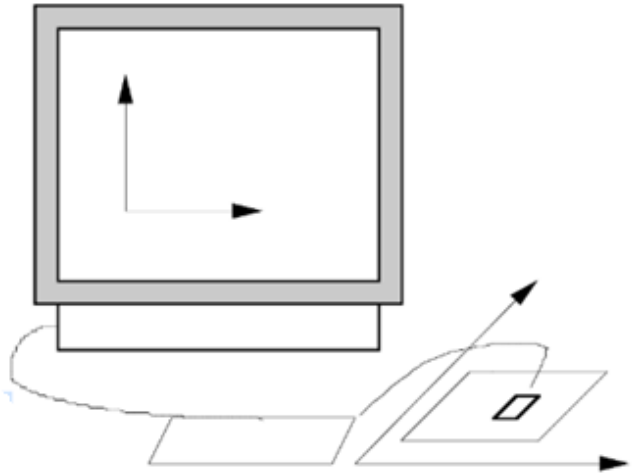
Major challenges:

- Natural language is vague & ambiguous
- Individual differences in speech
- Accents, emotions, illness affect recognition
- Background noise reduces accuracy

Examples :

- Voice assistants: “Hey Siri / Google” → commands, not long text
- WhatsApp voice messages: Faster than typing, but not searchable

Mouse



- Handheld pointing device
  very common
  easy to use

- Two characteristics

  planar movement
  buttons

- Mouse located on desktop requires physical space
- no arm fatigue
- Relative movement only is detectable.
- Screen cursor oriented in (x, y) plane,
- mouse movement in (x, z) plane

Mouse Cont.

- some experiments with the foot mouse (controlling mouse movement with feet)
- Why foot mouse failed: **Accidental activation.**
- Feet better suited for **simple, single-dimension controls**

Examples: car pedals, sewing machine speed control, gaming simulators, music production

Touchpad

- Touchpad = small, touch-sensitive input surface
- Commonly used in laptops (replaces mouse)
- Cursor is controlled by stroking a finger

good ‘acceleration’ settings important
fast stroke (lots of pixels per inch moved)
slow stroke (less pixels per inch for accurate positioning)

Trackball

- Trackball = upside-down mouse
  Ball is rotated by fingers
  Device itself stays fixed
- Indirect pointing device
  Controls cursor via relative motion
  Requires buttons for selection

In HCI, trackballs are used in space-constrained, high-precision, and accessibility-focused environments. They reduce physical strain, support rapid cursor movement, and are effective for specialized applications such as gaming, kiosks, and RSI-friendly systems.

Joysticks

- Joystick is an **indirect input device (**User controls a cursor via a control stick)
- **Two types (HCI distinction)**:
- **Absolute joystick**
  - Stick position ↔ cursor position
  - Used where spatial mapping matters
- **Isometric (velocity-controlled) joystick**
  - Stick pressure ↔ cursor speed
  - Stick returns to center when released

- Joysticks are effective for velocity-based control and navigation tasks, but are less suitable for high-precision pointing.

Question??

Kirsh Kumar is designing a desktop application for photo editing that requires selecting very small icons and dragging objects precisely. Which input device is more suitable: mouse or joystick? Why?

Response

A mouse is more suitable because it offers precise cursor control, which is required for selecting small icons and accurate dragging.

Question??

Laiba is creating a flight simulation game that requires continuous movement control. Which input device should be preferred and why?

Response

A joystick should be preferred because it supports continuous velocity-based control, ideal for navigation and simulation tasks.

Touchscreens

- Input + output device combined
- Detects finger or stylus touch directly on the screen
- A direct input device → no cursor–hand mapping needed

Advantages (HCI perspective)

- Very **fast interaction**
- **Intuitive** (point what you want)
- Works well in **public & harsh environments**

Disadvantages (HCI perspective)

- Low precision (finger is blunt)
- Difficult for **small targets & detailed drawing**

Stylus and light pen

Stylus:

- small pen-like pointer to draw directly on screen
- may use touch sensitive surface or magnetic detection
- used in PDA, tablets PCs and drawing tables

Light Pen:

- now rarely used
- uses light from screen to detect location

Stylus and light pens are direct input devices offering high precision but suffer from fatigue, occlusion, and handling issues, influencing screen layout design.

Digitizing Tablet

- Input device used mainly for:
  Freehand drawing
  Graphic design
  Map digitization
- Can also act as a mouse substitute
- Uses a stylus or puck on a special tablet surface
- Cursor position on screen maps to tablet position

A digitizing tablet is a high-precision indirect input device used mainly for drawing and design, offering accurate pen-based input at the cost of space and portability.

Eyegaze (Eye-Tracking Input)

- Input technique that controls the computer using eye
- Cursor or selection follows where the user looks
- Works by tracking retinal reflection using a low-power laser or camera

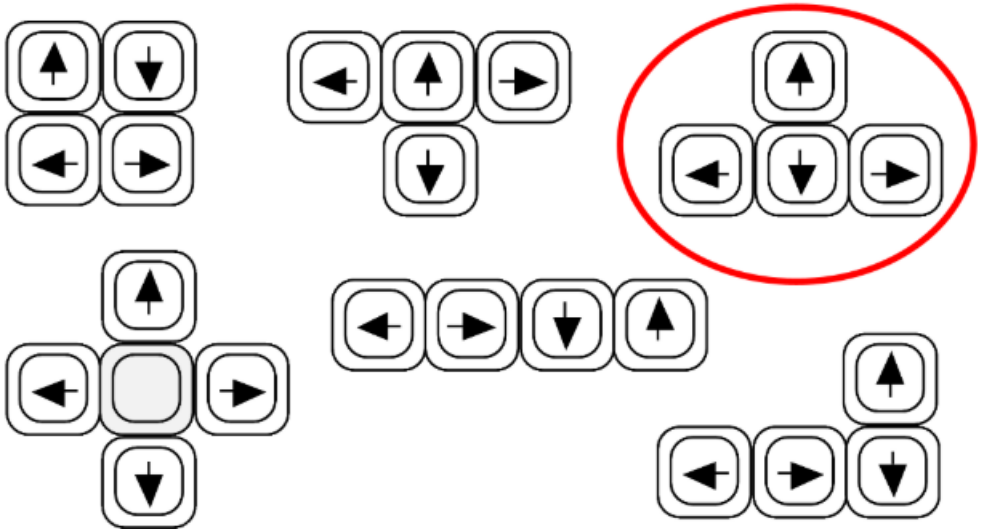
How It Works:

- User **calibrates** system by looking at dots on screen
- System tracks eye direction
- Cursor or action follows gaze direction

Eyegaze is powerful for fast selection and analysis of user attention but requires careful design to avoid accidental activation.

Cursor keys

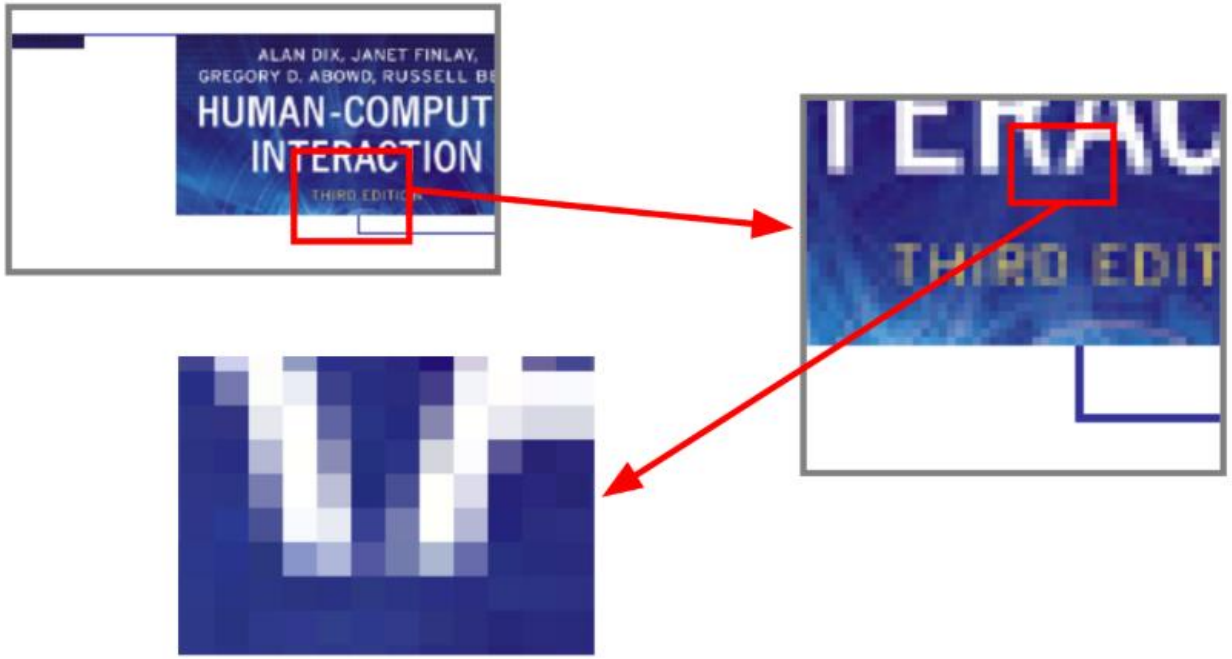
- Four keys (up, down, left, right) on keyboard.
- Very, very cheap, but slow.
- Useful for not much more than basic motion for text-editing tasks.
- No standardised layout, but inverted “T”, most common



Bitmap Displays - Resolution and Color

- The screen is made of tiny dots (pixels) arranged in a rectangular grid.
- Each pixel can display a color or intensity:
  1 bit/pixel → black & white
  8 bits/pixel → 256 colors
  24–32 bits/pixel → millions of colors (true color)

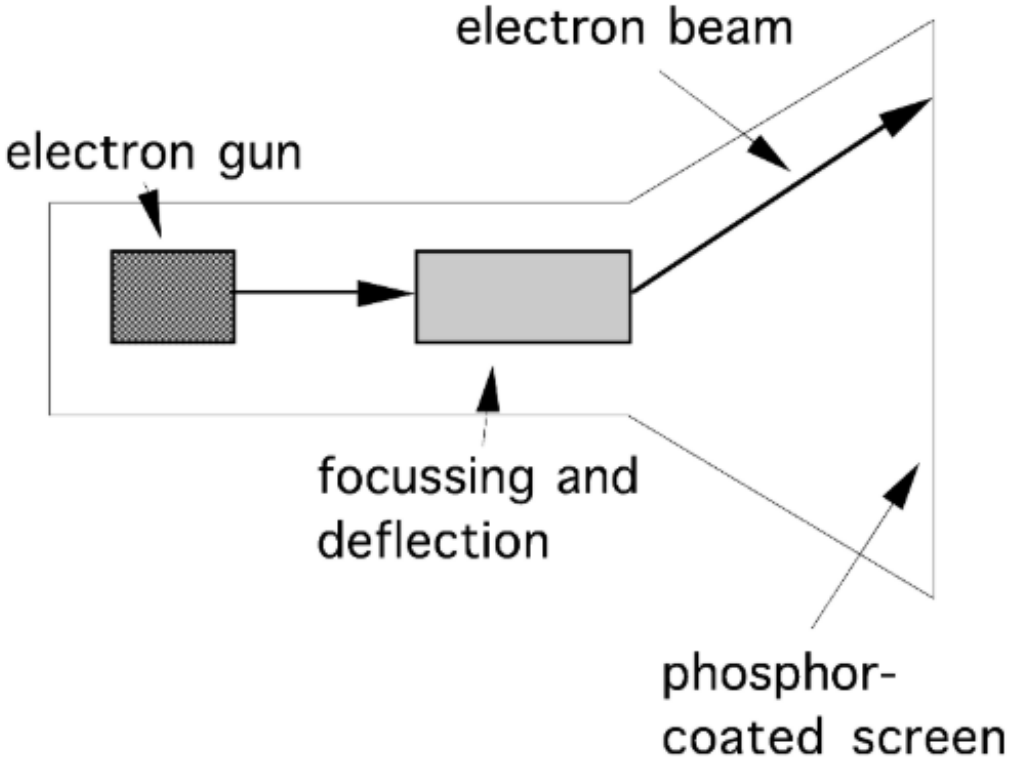
Pixels are the smallest controllable elements on a display. The computer's video card controls them.



Cathode Ray Tube

- Stream of electrons emitted from electron gun, focused and directed by magnetic fields, hit phosphor-coated screen which glows

- used in old TVs and computer monitors



Liquid Crystal Displays

- Smaller, lighter, and no radiation problems.
- Found on PDAs, portables and notebooks, and increasingly on desktop and even for home TV

  also used in dedicated displays digital watches, mobile phones, HiFi controls

How it works:

- Top plate transparent and polarized, bottom plate reflecting.
- Light passes through top plate and crystal and reflects back to eye.
- Voltage applied to crystal changes polarization and hence color
- N.B. light reflected not emitted => less eye strain

Special Displays

Gas Plasma Displays

- Used mainly for **large screens**, like big TVs or info boards.
- Bright, colorful, and visible from a distance.

Random Scan (Directed-beam refresh, vector display)

- draw the lines to be displayed directly
- lines need to be constantly redrawn
- rarely used except in special instruments

Direct view storage tube (DVST)

- Similar to random scan but persistent => no flicker
- Can be incrementally updated but not selectively erased
- Used in analogue storage oscilloscopes

Situated Displays

- displays in ‘public’ places
  large or small
  very public or for small group
- display only
  for information relevant to location
- or interactive
  use stylus, touch sensitive screen
- in all cases … the location matters
  meaning of information or interaction is related to the location

Hermes a situated display

small displays beside office doors

handwritten notes left using stylus

office owner reads notes using web interface

Positioning in 3D space

- In regular computer use, a mouse moves a cursor on a 2D screen: left/right and up/down (2 degrees of freedom).
- In virtual reality or 3D applications, you don’t just move left/right or up/down—you also move forward/backward and rotate objects.
- Modern VR/AR systems like **Meta Quest 3**, **PlayStation VR2**, or **Apple Vision Pro** allow you to **move and interact naturally in 3D worlds**.
- Not just walking forward/backward/sideways—users **look, lean, and rotate**, controlling both position and orientation.

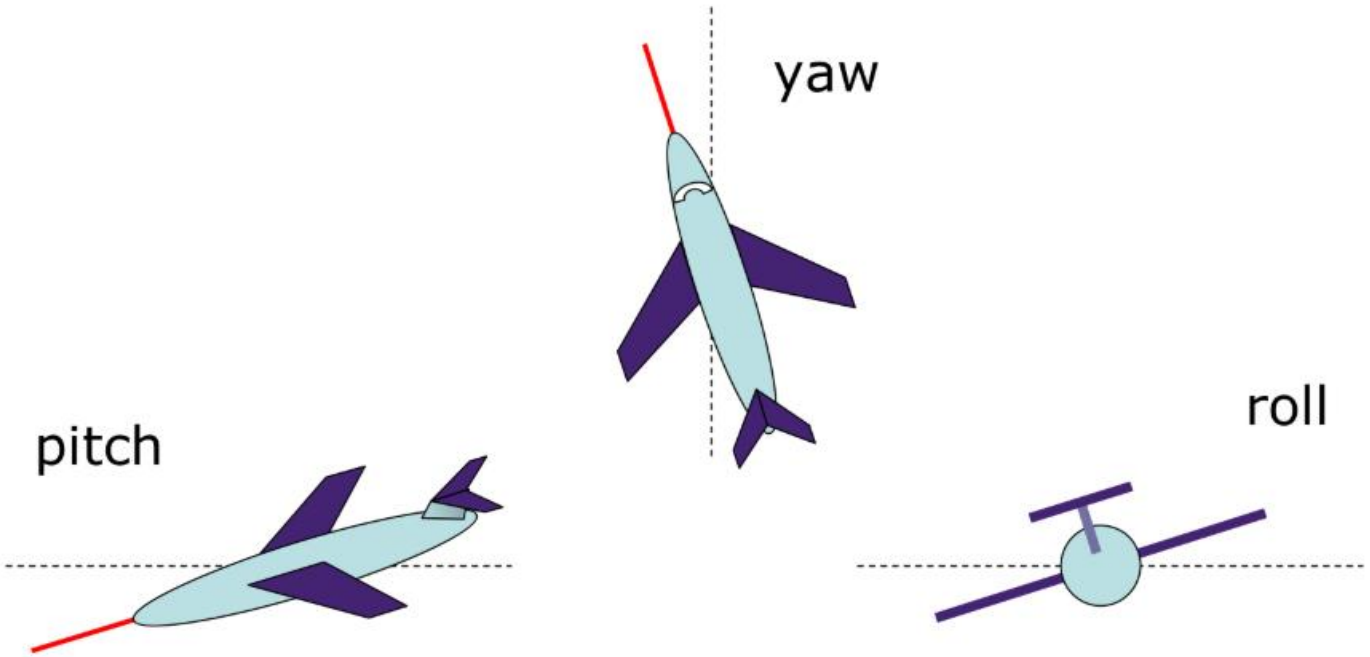
Cockpit and virtual controls

- steering wheels, knobs and dials.
- Flight simulators, racing games, or driving apps use **steering wheels, throttles, pedals, and knobs**.
- Example: **Microsoft Flight Simulator** or **Gran Turismo VR** uses realistic cockpit setups

3D Mouse / Spatial controllers

3Dconnexion SpaceMouse for CAD, or Meta Quest controllers for VR.

- Six degrees of freedom:
  X, Y, Z → position in 3D space
  Pitch, Yaw, Roll → orientation of device
- HCI Aspect: Direct mapping of physical movement to digital movement enhances precision and immersion, though fatigue is a concern for long sessions.



Data gloves

- Fiber optics or sensors track finger bending and hand gestures.
- HCI Aspect: Allows fine-grained manipulation, like picking up objects, drawing, or typing in VR. Feedback can include vibration or force (haptics) for realism.

VR helmets / Headsets

- Track head motion and sometimes eye gaze for selection or attention tracking.
- HCI Aspect: Eye tracking enables foveated rendering (reduces processing load where the user isn’t looking) and natural interaction with interfaces, making VR less tiring.

VR motion sickness

- time delay
  move head … lag … display moves
  conflict: head movement vs. eyes

- depth perception
  headset gives different stereo distance
  but all focused in same plane
  conflict: eye angle vs. focus

Special Displays

Special displays are **visual outputs other than the normal computer screen**.

- They are designed to show **critical information quickly and clearly**, often in systems where attention, safety, or speed matters more than rich graphics.
  1. LEDs (status indicators)
- **What they do:**
  Use simple lights to show system state.

Examples

- Laptop charging light (red = charging, green = full), Wi-Fi router blinking lights, PlayStation / Xbox power light

HCI aspect:

Quick feedback, No thinking required, Works even when screen is off

Dials and gauges

Show values using **analog visuals** instead of numbers.

Examples:

- Car speedometer and fuel gauge
- Bike RPM meter
- Smart AC temperature dial

Why not just numbers? (HCI)

- Humans spot **changes and extremes** faster on dials
- Easier to see “too high / too low” at a glance

Head-Up Display (HUD)

A display that shows information **directly in the user's line of sight**, so they don't need to look away.

Originally used in:

- Fighter aircraft
- Helicopters

Why pilots need HUDs (HCI reasoning)

Pilots must:

- Watch the sky
- Control the aircraft
- Monitor speed, altitude, fuel, threats
- Looking down at instruments = **danger**

Sound Output

Sound output means using **audio signals** (not speech alone) to communicate with the user or to give **feedback** about what the system is doing.

In HCI context, sound is mainly used to:

- Confirm an action
- Warn the user
- Reduce confusion
- Improve speed and confidence during interaction

Touch, feel, smell

Touch and feeling important

in games … vibration, force feedback

in simulation … feel of surgical instruments called haptic devices

texture, smell, taste

BMW iDrive: for controlling menus (feel small ‘bumps’ for each item)

makes it easier to select options by feel uses haptic technology from Immersion Corp.

Physical controls

- specialist controls needed … industrial controls, consumer products, etc.

Environment and Bio-sensing

Environment sensing enables interaction without explicit user controls
users interact naturally through presence or actions
Examples:
- Automatic washbasins using **infrared sensors** to detect hands
- Car **courtesy lights** triggered by door switches
- Such interactions are often **invisible,** and users may not notice the sensors

Common applications:
- Automatic doors
- Energy-saving motion-sensor lights
- Security tags in retail stores

Bio-Sensing and Behavioral Sensing

Sensors can measure a wide range of data:

- **Environmental**: temperature, movement, location (GPS), pressure
- **Behavioral**: audio, video, typing rate, vocabulary changes
- **Physiological**: body temperature, blink rate, heart rate

Examples:

- Smartwatches tracking stress or sleep
- Face unlock & emotion filters on phones
- Adaptive game difficulty based on player behavior

Printing

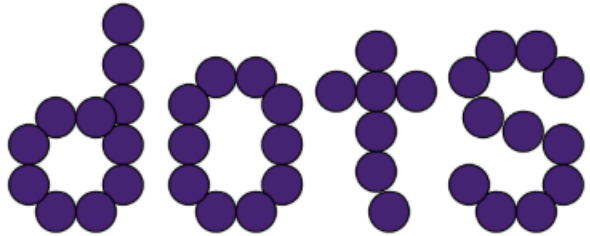


Image made from small dots

- allows any character set or graphic to be printed.

Resolution measured in **dots per inch (dpi)**

- Higher dpi → better visual quality
- Physical constraints affect perceived quality even at same dpi

Common dot-based printers:

- **Dot-matrix**: 80–120 dpi (low quality, noisy)
- **Ink-jet**: 300–600 dpi (photo-quality possible)
- **Laser**: 600–2400 dpi (high-quality text & graphics)

HCI Focus: Output quality affects **readability, satisfaction, and trust**

Fonts

Font – the particular style of text

Courier font
Helvetica font
Palatino font
Times Roman font

- ♣×∝≡↵ℜ€⊗↵~€ (special symbol)

Size of a font measured in points (1 pt about 1/72”)
(vaguely) related to its height

This is ten point Helvetica
This is twelve point
This is fourteen point
This is eighteen point
and this is twenty-four point

HCI Focus:

- Font choice affects **readability, aesthetics, and user experience.**
- Page description languages balance **performance, quality, and usability.**
- Consistent font rendering ensures predictable output across devices

Readability of text

- Lowercase: Easy to read shape of words
- UPPERCASE: better for individual letters and non-words e.g. flight numbers: BA793 vs. ba793
- Serif fonts: helps your eye on long lines of printed text but sans serif often better on screen

Scanners

- Take paper and convert it into a bitmap
- Two sorts of scanner
  flat-bed: paper placed on a glass plate, whole page converted into bitmap
  hand-held: scanner passed over paper, digitizing strip typically, 3-4” wide
- Shines light at paper and note intensity of reflection colour or greyscale
- Typical resolutions from 600–2400 dpi

OCR, Image Processing

- Converts scanned bitmap images into editable text
- Modern OCR can recognize unseen fonts, preserve layout (italics, subscripts, centering)
- Operates on bitmap images, not directly on optics

Rise of **digital cameras**: Direct digital capture → no scanning required

HCI Focus:

- Supporting search, retrieval, and reuse of information
- Interaction techniques for viewing and manipulating images
- Consistency of user experience across devices and media

Short-term Memory - RAM

- Random access memory (RAM)
  on silicon chips
  100 nano-second access time
  usually volatile (lose information if power turned off)
  data transferred at around 100 Mbytes/sec

- Some non-volatile RAM used to store basic set-up information

Long-Term Memory (LTM)

- **Long-Term Memory (LTM)** in computers stores information **persistently**
- Data remains even when power is off

Examples:

- Laptop internal storage (SSD/HDD)
- External hard drives/USB
- Cloud-backed local storage (Google Drive sync folders)
- Automatic version history in tools like Google Docs

Finite processing speed

Slow Processor – Overcomplicated Interfaces

- **Problem:** Designers assume processors are fast → interfaces get more complex.
  Issue: System cannot keep up with all tasks → lag and user frustration.

Example Icon wars: Clicking multiple icons or menu options in Windows/Mac → delayed response → windows/menu flash suddenly

Finite processing speed

Too-Fast Processor – Scrolling & Readability

- **Problem:** Systems may respond faster than the user can process.

Examples:

- Help screens scroll too quickly → text unreadable
- TikTok stories or Instagram Reels advance before you can read captions

Limitations on interactive performance

Interactive systems can be slowed by multiple factors.

- **Computation Bound** – delays due to heavy calculations

Example: Searching/replace in huge Google Docs

- **Storage/Memory Bound** – slow memory access

Example: Loading a compressed photo album on your phone

- **Graphics Bound** – rendering UI takes time

Example: Slow animations in mobile games

- **Network Bound** – delays in accessing remote data

Example: Buffering TikTok videos or Discord voice chat

Networked computing

Networks allow access to

- large memory and processing
- other people (groupware, email)
- shared resources – esp. the web

Issues

- network delays – slow feedback
- conflicts - many people update data

Class Activity 2

https://tinyurl.com/Spring2026-Activity2

Responses

1. Tourist Information System

Input device: Touchscreen

- **Reason:**
  Tourist information systems are typically used by walk-up users with little or no training. A touchscreen supports **direct manipulation**, which is a key. Users can easily tap icons, scroll maps, and select options without needing instructions or prior computer experience.

Output device: Large visual display

- **Reason:**
  Tourists benefit most from **visual output** such as maps, images, directions, and symbols. A large screen allows information to be presented clearly and quickly, overcoming language barriers and supporting fast understanding.

Responses

Air Traffic Control System

Input Device: Light Pen / Stylus

- **Reason:**
  An air traffic control system is a **safety-critical, real-time system** where controllers must interact quickly and accurately with aircraft symbols on radar screens. A light pen or stylus provides **high-precision direct interaction**, allowing controllers to point directly at aircraft, routes, or conflict areas on the display.

Output device: High-resolution radar display

- **Reason:**
  Controllers must maintain **continuous situational awareness**. A high-resolution visual display shows aircraft positions, altitude, speed, and trajectories in real time, supporting accurate monitoring and decision-making.

Responses

Virtual Reality (VR) Training System

Input Device: Motion Sensors / Hand Controllers

- **Reason:**
  VR training systems require users to interact in a **3D virtual environment**. Motion sensors capture body and hand movements, enabling: Realistic interaction
- Skill-based training (e.g., surgery, equipment handling)
- Compared to keyboard or mouse, motion input provides **natural mapping** between real-world actions and virtual actions.

Output Device: Head-Mounted Display (HMD)

- **Reason:**
  An HMD provides immersive visual output, allowing users to experience realistic environments. This improves **learning effectiveness and engagement** by simulating real-world conditions.